Dual-ligand screen / Setup

48 pairs, then 22 pairs + L17 reference. One 48 × 2 mL Para-Dox, 71 fresh reaction vials. Manual caps and labware moves. **All precise liquid handling is robotic**, including stocks, standards, R2 predoses and source refills.

Load chemicals

- Weigh labeled charges from `weighed_charges_template.csv`; enter actual masses/assays. Weigh ligands in the glovebox. Robot adjusts solvent addition; respect mass limits.
- Coarse-load raw solvents to CSV minimum fills. Supplied **8 ×30 mL A1-H1 reservoir: single-channel ONLY**. No manual aliquoting or volumetric top-up.
- Bulk carriers use slots **3/6** by ID. Confirm campaign-length solvent compatibility of provisional 15 mL conical containers, or supply exact compatible glass-source JSON. Retain capped reserves and original working tubes.
- Have **three compatible 96-well LC plates**, 2.2 mL geometry, 1 mL/sample, carrier/seals. Confirm offsets, mounting, caps and actual 1.7-min method/cycle. Import/simulate locally.

Pipettes

Initial preparation: p300 single left /p1000 single right; 300 µL tips in 5, 1000 µL in 8/9. After preparation, change RIGHT to **p300 eight-channel GEN2** and tips 8/9 to 300 µL via the App installation/calibration workflow. Left stays installed.

Stocks - all liquids prepared by robot

Stock	Rule
Ligands	Nominal **20 mM**, generous reserves. Weighed charges in mapped carriers.
RockPhos L03	**~7.50 mg pure equivalent**, assay-corrected; 10 mg available. Robot makes **~800 µL in 1 C1**. Retain SAME tube; **max 4 R2 wells**.
Pd(COD)(DQ)	**90 mg maximum**, nominal **40 mM /~5.94 mL**. SAME bulk both rounds; **10 mg solid reserved for scope**.
SM + mesitylene	**~322.32 mg SM** at 100% assay; robot adds neat IS/toluene: nominal **1 M /100 mM**, ~2 mL. **4 D2: 1.88 mL** for all 71 vials.

Concentrations use nominal solvent addition. **No hand refills or R2 stock remake.**

Reaction-stage deck

Slot	Contents
1 /4	2 mL working tubes, max 1.95 mL. Neat IS **1 C2**, SM/IS **4 D2**, Pd **4 A3/B3**.
2	EMPTY 12-channel reservoir; robot makes quench A4/A6/A9 and diluent A5/A7/A10/A11/A12.
3 /10	Reaction block: pipetting /heater-shaker. During holds, raw solvents temporarily occupy 3.
4 /6 in R1 hold	Prepared analytical rack temporarily in 4; calibration plate in 6, then directly to LC. Later, 6 takes each final LC plate.
5 /8 /9	Tips. Keep **7 and 11 empty**.

Dual-ligand screen / Run

Operator / start: _____ LC method / channel: _____

Follow App pauses and matching loading CSVs. Staff handle weighing, coarse liquid loading, caps, rack/plate moves and local starts. Robot preparation counts inside the campaign.

Step	Action
1 / Stocks	Run `prepare_stocks.py` from actual weights. Load requested carriers uncapped; cap/store when removed. Retain the preparation manifest. Make the one right-pipette/tip-rack change after preparation.
2 / R2 predose first	Load **23 NEW R2 vials** in the SAME reaction block. R1 Python doses **25 µL SM/IS each**. Hand-cap/store them, then fit **48 NEW EMPTY R1 vials** and uncap. No manual pipetting.
3 / R1 reaction	Robot doses **25 µL SM/IS +75 µL ligand A +75 µL ligand B +75 µL Pd**. Hand-cap; move/latch. **95 °C, 500 rpm, 4 h.**
4 / During R1 hold	Leave reaction on 10. At prompts, load raw solvents in 3 and analytical rack in 4. Robot makes **14 calibration standards**, pipette-mixes them and prepares quench/diluent in 2. Move sealed calibration plate **6 → HPLC** and start its sequence. Remove raw reservoir from 3 and restore working rack 4.
5 / Cool / quench	Move sealed block to 3; cool before opening. Robot adds **500 µL quench**. Hand-recap ALL; shake **1500 rpm, 2 min**, heater off; return to 3.
6 / LC samples	Let bubbles clear; uncap indicated vials. Robot adds **20 µL extract to 980 µL diluent**, fresh sample tips. Recap reaction vials.
7 / Final plate	Move **6 → cooled shaker 10**. Robot shakes **UNCOVERED, heater OFF, 1000 rpm, 2 min** at confirmed splash-free speed. Move **10 → HPLC**, seal and inject directly. No manual dilution/vial transfers.
8 / R2 select/refill	After all **48 valid R1 results**, Claude selects **22 new pairs +L17 reference**. Run `refill_stocks.py` **FIRST**: present original bulk carriers; robot tops up original working tubes. Retain RockPhos.
9 / R2	Restore the SAME block with 23 capped robot-predosed vials. Replace reservoir 2 with a NEW EMPTY one. Uncap all 23; run R2 Python. **No SM/IS redose.** Reference uses **150 µL L17**, no B. Repeat reaction/workup; robot prepares workup fluids during this hold too. **R2 H3 is diluent only: do not inject.**

Calibration: each analyte blank +**10, 50, 150, 350, 800 µM**, independent **666.7 µM** check; constant **66.7 µM mesitylene**. Python calculates calibrated yields from integrated UV areas. Stop the clock after all 23 final injections and quantitative analysis.

Pause for blocked tips, persistent bubbles, air uptake, obvious layers or tip contact. Do not guess liquid top-ups or repair a dilution by hand. Report deviations so stock accounting can be corrected.

Timing target: two 4 h holds plus approximately **4 h for remaining tasks, including LC**. Record actual times; the target is advisory. Give logs/raw LC files/stock and deviation records to the pipeline operator; return them using `python scripts/hte-return-package.py`.